

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think> Reasoning process :

- 1 . The anatomical snuff box is a region in the wrist that is located on the radial , dorsal aspect of the hand .
- 2 . The anatomical snuff box is bordered by the tend ons of the ext ensor pollic is long us , ext ensor car pi radialis bre vis , and ext ensor car pi radialis long us muscles .
- 3 . The radial artery is a major artery that runs through the anatomical snuff box .
- 4 . The radial artery is more superficial and is located in the anatomical snuff box , making it more susceptible to injury .
- 5 . The ul nar artery is located more med ially and is not typically associated with the anatomical snuff box .
- 6 . The Prince ps pollic is artery is a branch of the radial artery and is located in the first met ac arp al space , but it is not the main artery responsible for bleeding in the anatomical snuff box .
- 7 . The brachial artery is a larger artery that runs through the upper arm and is not directly associated with the anatomical snuff box .

</think>< answer >C . Prince ps pollic is artery is not the main artery , the correct answer is A . Radial artery </answer >